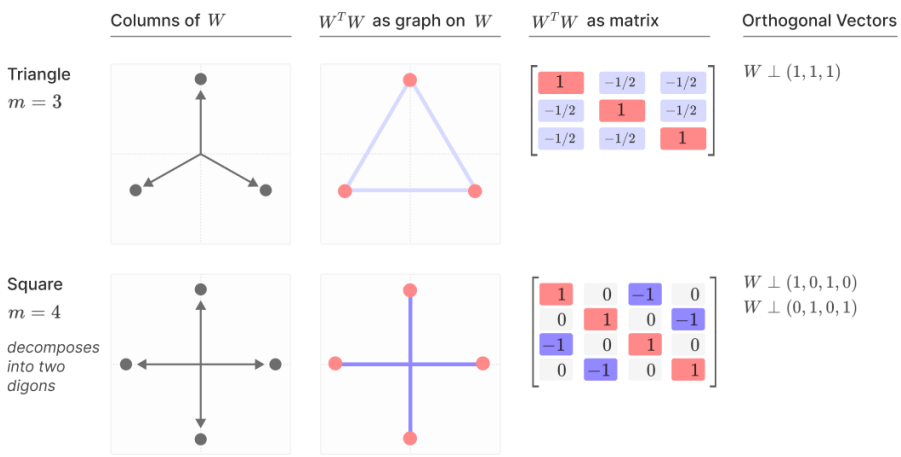


This correspondence also goes the other direction. Suppose we have a rank $(n-i)$ -matrix of the form W^TW. We can characterize it by the dimensions W *did not* represent – that is, which directions are orthogonal to W? For example, if we have a $(n-1)$ -matrix, we might ask what single direction did W not represent? This is especially informative if we assume that W^TW will be as "identity-like" as possible, given the constraint of not representing certain vectors.

In fact, given such a set of orthogonal vectors, we can construct a polytope by starting with n basis vectors and projecting them to a space orthogonal to the given vectors. For example, if we start in three dimensions and then project such that $W \perp (1,1,1)$, we get a triangle. More generally, setting $W \perp (1,1,1,...)$ gives us a regular n-simplex. This is interesting because it's in some sense the "minimal possible superposition." Assuming that features are equally important and sparse, the best possible direction to not represent is the fully dense vector $(1,1,1,...)$!

Non-Uniform Superposition

So far, this section has focused on the geometry of uniform superposition, where all features are of equal importance, equal sparsity, and independent. The model is essentially solving a variant of the Thomson problem. Because all features are the same, solutions corresponding to uniform polyhedra get especially low loss. In this subsection, we'll study non-uniform superposition, where features are somehow not uniform. They may vary in importance and sparsity, or have a correlational structure that makes them not independent. This distorts the uniform geometry we saw earlier.

In practice, it seems like superposition in real neural networks will be non-uniform, so developing an understanding of it seems important. Unfortunately, we're far from a comprehensive theory of the geometry of non-uniform superposition at this point. As a result, the goal of this section will merely be to highlight some of the more striking phenomena we observe:

- **Features varying in importance or sparsity** causes smooth deformation of polytopes as the imbalance builds, up until a critical breaking point at which they snap to another polytope.
- **Correlated features** prefer to be orthogonal, often forming in different tegum factors. As a result, correlated features may form an orthogonal local basis. When they can't be orthogonal, they prefer to be side-by-side. In some cases correlated features merge into a single feature: this hints at some kind of interaction between "superposition-like behavior" and "PCA-like behavior".
- **Anti-correlated features** prefer to be in the same tegum factor when superposition is necessary. They prefer to have negative interference, ideally being antipodal.

We attempt to illustrate these phenomena with some representative experiments below.

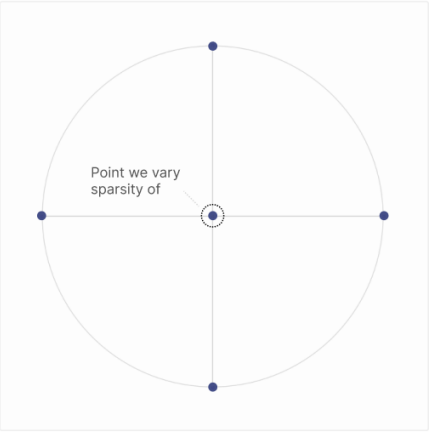
PERTURBING A SINGLE FEATURE

The simplest kind of non-uniform superposition is to vary one feature and leave the others uniform. As an experiment, let's consider an experiment where we represent $n = 5$ features in $m = 2$ dimensions. In the uniform case, with importance $I = 1$ and activation density $1 - S = 0.05$, we get a regular pentagon. But if we vary one point – in this case we'll make it more or less sparse – we see the pentagram *stretch* to account for the new value. If we make it denser, activating more frequently (yellow) the other features repel from it, giving it more space. On the other hand, if we make it sparser, activating less frequently (blue) it takes less space and other points push towards it.

If we make it sufficiently sparse, there's a phase change, and it collapses from a pentagon to a pair of digons with the sparser point at zero. The phase change corresponds to loss curves corresponding to the two different geometries crossing over. (This observation allows us to directly confirm that it is genuinely a first order phase change.)

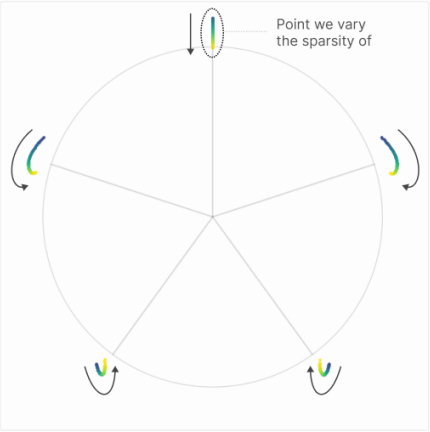
To visualize the solutions, we canonicalize them, rotating them to align with each other in a consistent manner.

Digon (Square) Solutions



When the sparsity of the varied point falls below a certain critical threshold (~2.5x less than others) the pentagon solution changes to two digons.

Pentagon Solutions



Note how vertices shift as sparsity changes

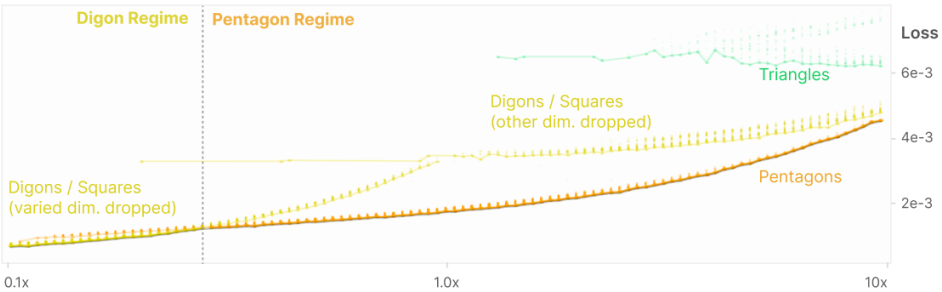
To study non-uniform sparsity, we consider models with five features, varying the sparsity of a single feature and observing how the resulting solutions change. We observe a mixture of continuous deformation and sharp phase changes.

Parameters

n = 5
m = 2
I_i = 1
$1\text{-}S$ = 0.05 (baseline)

Relative Feature Density (1-S)

The Pentagon-Digon Phase Change Corresponds to a Loss Curve Crossover



Gradient descent has trouble moving between solutions associated with different geometries. As a result, fitting the model will often produce non-optimal solutions. By characterizing and plotting these, we can see that each geometry creates a different loss curve, and that the pentagon-digon phase change corresponds to a cross over between the curves.

These results seem to suggest that, at least in some cases, non-uniform superposition can be understood as a *deformation of uniform superposition* and *jumping between uniform superposition configurations* rather than a totally different regime. Since uniform superposition has a lot of understandable structure, but real world superposition is almost certainly non-uniform, this seems very promising!

The reason pentagonal solutions are not on the unit circle is because models reduce the effect of positive interference, setting a slight negative bias to cut off noise and setting their weights to $||W_i|| = 1/(1-b_i)$ to compensate. Distance from the unit circle can be interpreted as primarily driven by the amount of positive interference.

A note for reimplementations: optimizing with a two-dimensional hidden space makes this easier to study, but the actual optimization process to be really challenging from gradient descent – a lot harder than even just having three dimensions. Getting clean results required fitting each model multiple times and taking the solution with the lowest loss. However, there's a silver lining to this: visualizing the sub-optimal solutions on a scatter plot as above allows us to see the loss curves for different geometries and gain greater insight into the phase change.

Correlated and Anticorrelated Features

A more complicated form of non-uniform superposition occurs when there are correlations between features. This seems essential for understanding superposition in the real world, where many features are correlated or anti-correlated.

For example, one very pragmatic question to ask is whether we should expect polysemantic neurons to group the same features together across models. If the groupings were random, you could use this to detect polysemantic neurons, by comparing across models! However, we'll see that correlational structure strongly influences which features are grouped together in superposition.

The behavior seems to be quite nuanced, with a kind of "order of preferences" for how correlated features behave in superposition. The model ideally represents correlated features orthogonally, in separate tegum factors with no interactions between them. When that fails, it prefers to arrange them so that they're as close together as possible – it prefers positive interference between correlated features over negative interference. Finally, when there isn't enough space to represent all the correlated features, it will collapse them and represent their principal component instead! Conversely, when features are anti-correlated, models prefer to have them interfere, especially with negative interference. We'll demonstrate this with a few experiments below.

SETUP FOR EXPLORING CORRELATED AND ANTICORRELATED FEATURES

Throughout this section we'll refer to "correlated feature sets" and "anticorrelated feature sets".

Correlated Feature Sets. Our correlated feature sets can be thought of as "bundles" of co-occurring features. One can imagine a highly idealized version of what might happen in an image classifier: there could be a bundle of features used to identify animals (fur, ears, eyes) and another bundle used to identify buildings (corners, windows, doors). Features from one of these bundles are likely to appear together. Mathematically, we represent this by linking the choice of whether all the features in a correlated feature set are zero or not together. Recall that we originally defined our synthetic distribution to have features be zero with probability S and otherwise uniformly distributed between [0,1]. We simply have the same sample determine whether they're zero.

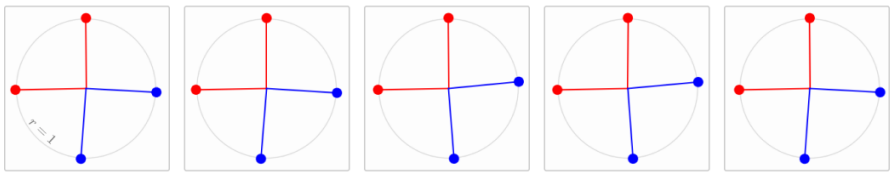
Anticorrelated Feature Sets. One could also imagine anticorrelated features which are extremely unlikely to occur together. To simulate these, we'll have anticorrelated feature sets where only one feature in the set can be active at a time. To simulate this, we'll have the feature set be entirely zero with probability S, but then only have one randomly selected feature in the set be uniformly sampled from [0,1] if it's active, with the others being zero.

ORGANIZATION OF CORRELATED AND ANTICORRELATED FEATURES

For our initial investigation, we simply train a number of small toy models with correlated and anti-correlated features and observe what happens. To make this easy to study, we limit ourselves to the $m = 2$ case where we can explicitly visualize the weights as points in 2D space. In general, such solutions can be understood as a collection of points on a unit circle. To make solutions easy to compare, we rotate and flip solutions to align with each other.

Models prefer to represent correlated features in orthogonal dimensions.

We train several models with 2 sets of 2 correlated features (n=4 total) and a m=2 hidden dimensions. We then visualize the weight column for each feature. For ease of comparison, we rotate and flip solutions to have a consistent orientation.

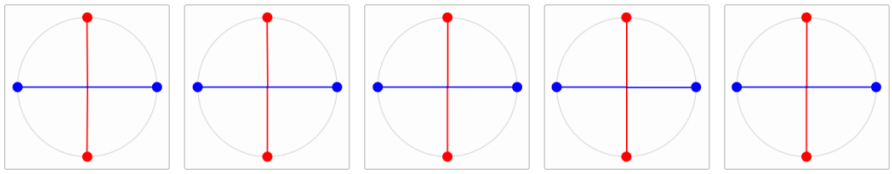


●● and ●● denote **correlated** feature sets.

Correlated feature sets are constructed by having them always co-occur (ie. be zero or not) at the same time.

Models prefer to represent anticorrelated features in opposite directions.

We train several models with 2 sets of 2 anticorrelated features (n=4 total) and a m=2 hidden dimensions. We then visualize the weight column for each feature. For ease of comparison, we rotate and flip solutions to have a consistent orientation.

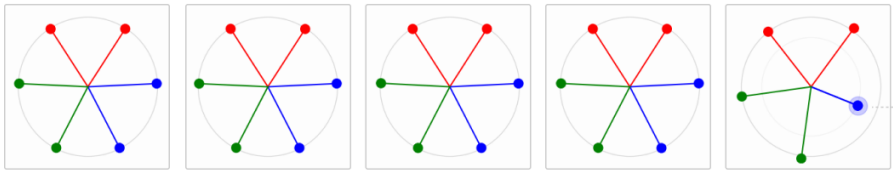


●● and ●● denote **anticorrelated** feature sets.

Anticorrelated feature sets are constructed by having them never co-occur (ie. be zero or not) at the same time.

Models prefer to arrange correlated features side by side if they can't be orthogonal.

We train several models with 3 sets of 2 correlated features (n=6 total) and a m=2 hidden dimensions. We then visualize the weight column for each feature. For ease of comparison, we rotate and flip solutions to have a consistent orientation. (Note that models will not embed 6 independent features as a hexagon like this.)



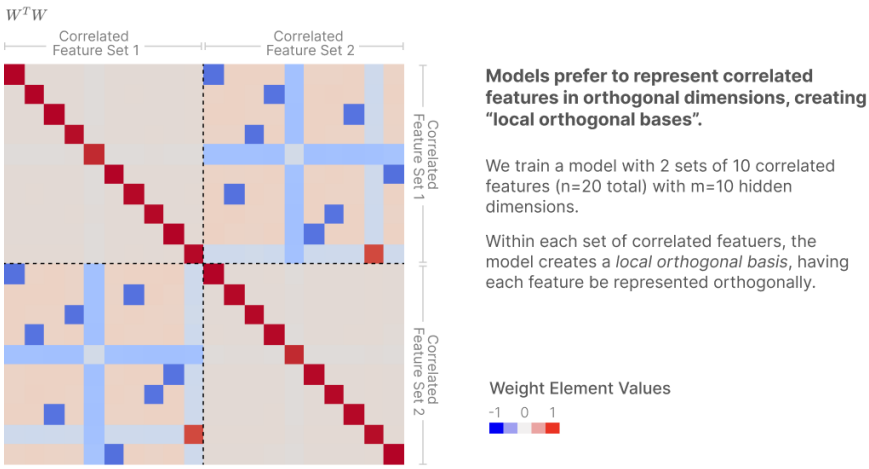
●●, ●●, and ●● denote **correlated** feature sets.

Sometimes correlated feature sets "collapse". In this case it's an optimization failure, but we'll return to it shortly as an important phennomenon.

LOCAL ALMOST-ORTHOGONAL BASES

It turns out that the tendency of models to arrange correlated features to be orthogonal is actually quite a strong phenomenon. In particular, for larger models, it seems to generate a kind of "local almost-orthogonal basis" where, even though the model as a whole is in superposition, the correlated feature sets considered in isolation are (nearly) orthogonal and can be understood as having very little superposition.

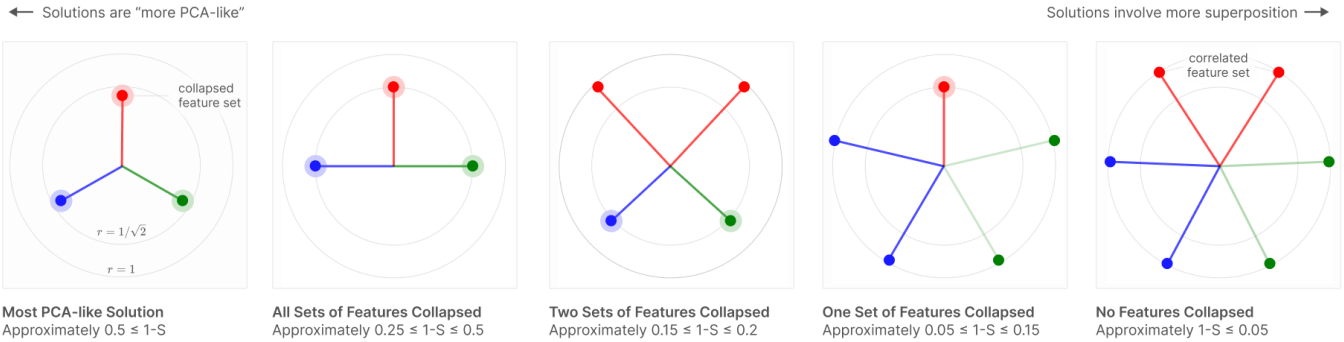
To investigate this, we train a larger model with two sets of correlated features and visualize W^TW.



If this result holds in real neural networks, it suggests we might be able to make a kind of "local non-superposition" assumption, where for certain sub-distributions we can assume that the activating features are not in superposition. This could be a powerful result, allowing us to confidently use methods such as PCA which might not be principled to generally use in the context of superposition.

COLLAPSING OF CORRELATED FEATURES

One of the most interesting properties is that there seems to be a trade off with Principal Components Analysis (PCA) and superposition. If there are two correlated features a and b, but the model only has capacity to represent one, the model will represent their principal component $(a+b)/\sqrt{2}$, a sparse variable that has more impact on the loss than either individually, and ignore the second principal component $(a-b)/\sqrt{2}$.

As an experiment, we consider six features, organized into three sets of correlated pairs. Features in each correlated pair are represented by a given color (red, green, and blue). The correlation is created by having both features always activate together – they're either both zero or neither zero. (The exact non-zero values they take when they activate is uncorrelated.)

As we vary the sparsity of the features, we find that in the very sparse regime, we observe superposition as expected, with features arranged in a hexagon and correlated features side-by-side. As we decrease sparsity, the features progressively "collapse" into their principal components. In very dense regimes, the solution becomes equivalent to PCA.